

Phase 7

FitBuddy- AI Fitness Plan Generator using Gemini Models

Project Documentation

1. Introduction

Documentation is an essential phase in the development of the **FitBuddy – AI Fitness Plan Generator**. After completing the brainstorming, requirement analysis, project design, planning, development, and testing phases, the work carried out during the project must be organized into a structured document.

The purpose of project documentation is to provide a clear explanation of what FitBuddy is, why it was developed, how it works, what technologies are used, how the system was implemented, and how the application can be executed and demonstrated.

FitBuddy is documented as a web-based application that uses generative AI to create personalized workout plans and nutrition or recovery tips based on user information such as age, weight, fitness goal, and workout intensity. Users can also submit feedback and request an updated workout plan.

The documentation phase therefore brings together the technical and functional information produced throughout the project.

2. Purpose of Project Documentation

The main purpose of documenting FitBuddy is to make the project understandable, reproducible, maintainable, and presentable.

The documentation should explain:

- The project concept.
- The problem addressed by FitBuddy.
- The objectives of the application.
- User scenarios.
- Functional requirements.
- Technical architecture.
- Technologies used.
- AI model integration.
- Database implementation.
- Backend routing.
- Frontend implementation.

- Development workflow.
- Testing approach.
- Local deployment.
- User interaction flow.
- Project limitations.
- Future development possibilities.
- Demonstration procedure.

Good documentation also helps developers understand the relationship between different components of the application.

3. Documentation Objectives

The documentation phase has the following objectives.

Objective 1 – Explain the Project

Provide a clear description of FitBuddy and its purpose.

Objective 2 – Record Development Activities

Document the major activities completed during development.

Objective 3 – Explain Technical Architecture

Describe the relationship between the frontend, backend, AI layer, and database.

Objective 4 – Document Functionalities

Explain how workout generation, nutrition guidance, feedback, and user management work.

Objective 5 – Provide Execution Information

Document the environment setup and local execution process.

Objective 6 – Support Project Demonstration

Provide enough information to explain the application during a project presentation or demonstration.

Objective 7 – Support Future Maintenance

Allow another developer or student to understand the structure and operation of the project.

4. Project Documentation Structure

The FitBuddy documentation can be organized into the following major sections:

FitBuddy Project Documentation

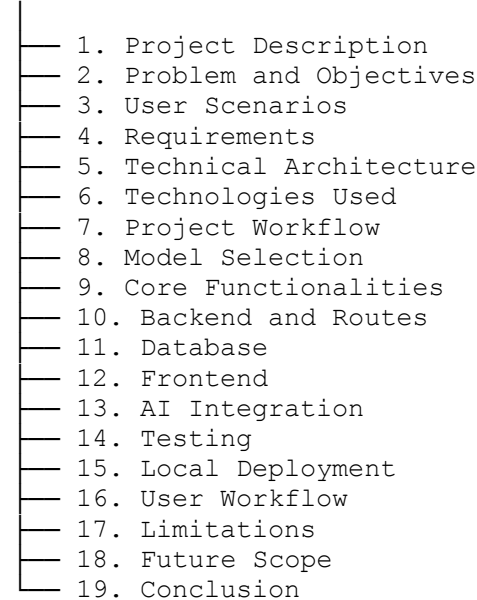


Figure 1: FitBuddy Project Documentation Structure

This structure provides a logical progression from the project concept to implementation and final deployment.

5. Project Description

FitBuddy is described in the source project as an **AI-powered Fitness Plan Generator**.

The application uses Google's Gemini models to generate personalized workout plans and nutrition tips based on user fitness goals. The user provides information such as age, weight, fitness goal, and workout intensity, after which the application generates a customized seven-day fitness plan.

The system also supports a feedback mechanism. A user can submit feedback regarding an existing workout plan, and the application can use that feedback to generate an updated plan.

FitBuddy uses:

- FastAPI for backend processing.
- Google Gemini for generative AI.
- SQLite for persistent storage.
- SQLAlchemy f